

Grow & Care

Stage 2: User and task analysis

Problem

Parents and caregivers often struggle to find reliable, age-appropriate, and personalized activities that support their children's development while still fitting into busy routines. Existing resources are scattered and time-consuming to evaluate, making it difficult to identify the best information and activities. At the same time, children need simple, enjoyable, while also safe digital experiences that promote learning and creativity. Grow & Care addresses this gap by providing a centralized, trustworthy platform that offers personalized activity and event suggestions, practical guidance, and a safe "Child Mode" for supervised play and learning.

Users

User Class 1: Parents and Caregivers (logged)

Characteristics:

Adults aged 25–50 who are responsible for children aged 3–12. They may include working parents with limited time, stay-at-home caregivers, or family members like grandparents. They often juggle multiple responsibilities and may struggle to find reliable, age-appropriate activities for their children.

Needs:

- Quick and trustworthy recommendations for activities.
- Ability to filter by age, developmental goals (e.g. reading, creativity, focus), time availability, and type (indoor, outdoor, online, local).
- Access to structured plans, local events, and educational materials.
- Practical guidance and safety tips, especially for new or less experienced parents.

Technical Skills:

Intermediate level overall. Most are comfortable with websites and mobile apps but prefer intuitive and simple interfaces that save time.

Motivation:

They want to support their children's development efficiently, reduce stress, save time, and feel confident in their choices. They value reliable content and clear guidance.

User Class 2: Parents and Caregivers (non-logged)

Characteristics:

Same as class 1, but they are only browsing the site and have not created a profile yet.

Needs:

- Quick overview of what the website has to offer
- Access to educational materials, practical guidance and safety tips

Technical Skills:

Same as class 1.

Motivation:

Browsing the site to see if it will help them out with their children and reading the educational articles/info texts.

User Class 3: Children (Limited Users)

Characteristics:

Children aged 3–12 who interact directly with the platform only in “Child Mode”, accessing selected online interactive materials (e.g. short stories, games, learning exercises). They use the website under the supervision of an adult.

Needs:

- Engaging and fun content suited to their age and interests.
- Simple, visual, and safe interface with minimal reading required.
- Activities that promote curiosity, play, and learning.

Technical Skills:

Very limited; they rely on intuitive visual navigation, icons, and audio prompts.

Motivation:

Play, curiosity, and enjoyment. Their engagement and preferences indirectly help caregivers refine activity choices.

Role:

Secondary users. While they do not manage the platform, their direct experience with activities (fun, engagement, comprehension) provides valuable feedback for parents and caregivers.

Tasks

The Grow&Care platform supports two main user groups:

- **Primary users:** logged in parents & caregivers (including educators) / browsing parents & caregivers
- **Secondary users:** children (in supervised “child mode”)

Presented below are the primary high-level tasks that each group is capable of performing.

Task 1: Search and Explore Activities

Objective: Allow users to explore different types of activities suited to their children’s needs.

Description: Users can browse or filter activities based on various criteria, including age, interests, developmental goals, or format.

Pre-condition: The user knows the child's age and interests.

Subtasks:

- Access the "Activities" section.
- Apply filters such as age range, indoor/outdoor, topic (creativity, reading, focus), or type (online/offline).
- Review suggested activities
- Open detailed descriptions to view requirements, benefits, and instructions.

Exceptions:

- No activities found matching criteria.
- Incomplete or unclear activity descriptions.
- Incorrect filter combinations.

Task 2: Receive Personalized Recommendations

Objective: Provide tailored suggestions based on the specific developmental needs and preferences of the child.

Description: The system generates recommendations after users input, parents and children have profiles

Pre-conditions: User has entered child's age, preferences, developmental goals, etc.)

Subtasks:

- Create or update a child's profile
- Select developmental goals (e.g. improve reading, boost concentration).
- Indicate preferences (e.g. outdoor vs indoor, duration).
- View personalized suggestions.

Exceptions:

- Missing or outdated profile data.
- Recommendations not relevant or too generic.

Task 3: Plan, Save, and Book Activities

Objective: Help users organize and manage selected activities, both online and offline.

Description: After finding or receiving suggestions, users can save them, plan when to do them, or book a spot for local events.

Pre-conditions: Activities are available; user is logged in into the platform.

Subtasks:

- Save or bookmark activities for later.
- Access a personal list or calendar.

- Book participation in workshops, events, or classes.

Exceptions:

- Event fully booked.
- Booking activities happening at the same time

Task 4: Access Guidance, Help, and Safety Tips

Objective: Support parents and caregivers with practical advice and reliable information that should be based on reliable information such as academic papers.

Description: Users can consult educational content, parenting tips, or safety guidelines (e.g. for newborn care or child development milestones).

Pre-conditions: None (open access).

Subtasks:

1. Navigate to the “Guidance” or “Tips” section.
2. Select topic (e.g. newborn safety, learning at home).
3. Read or save articles.

Exceptions: Content too general or not localized.

Task 5: Use Interactive Online Materials (Child Mode)

Objective: Enable children to safely engage with educational content in a simplified environment.

Description: In “Child Mode,” children can access selected online games, interactive stories, or creative exercises designed for their age.

Pre-conditions: “Child mode” selected by parent

Subtasks:

- Parent chooses age-range (& other prerequisites)
- Parent enables “Child Mode.”
- Child chooses a game or story from a visual menu.
Completes the interactive activity

Exceptions:

- Task can be too challenging for the child.

Scenarios

Scenario 1: Exploring Activities (Non-Logged User)

Anna, mother of a 6-year-old, visits Grow&Care for the first time. She filters activities by age, indoor setting, and creativity. A list of craft ideas appears, each with brief descriptions and reviews. When she tries to save one, the system prompts her to register, explaining that only logged users can store favorites or receive personalized suggestions.

Scenario 2: Personalized Recommendations and Booking (Logged User)

After registering, Anna creates a profile for her daughter and selects the goal “improve focus.” Grow&Care suggests online games and local workshops. She books an art class directly through the website and it’s added to her saved list.

Scenario 3: Accessing Guidance and Tips

Later, Anna browses the Guidance & Tips section to read about creative learning and safety advice for young children. Articles are freely accessible and can be bookmarked for later reading, helping her stay informed without navigating external sites.

Scenario 4: Using Child Mode

Anna activates Child Mode so her daughter can safely play an interactive drawing game. To enter and exit this mode, she sets up a safety PIN, ensuring that her daughter stays within the restricted, age-appropriate environment. The interface is simplified and colorful, allowing independent, safe exploration.

Scenario 5: Child Mode used by a Child

After his mother activated Child Mode, 6 year-old Tom explores the games available to him. Drawn by the colorful icon, he clicks on an interactive word matching game. The game is self-explanatory, simple and engaging. After he is done, he tries to navigate to another page of the website, but the system notifies him that he cannot leave child mode without his mother entering the correct PIN. He hands the device over to his mother.

Interviews - Summary of Findings

Four semi-structured interviews were conducted with parents of children aged 1 to 12 to understand their routines, needs, and expectations for a platform like Grow&Care. Two conversations (informal interviews) with children were also included.

Giorgia, a 34-year-old mother from Rome (Italy) with three children aged 1, 4, and 6, manages a very busy household with support from her mother and a nanny. Her children enjoy outdoor play, dancing, baking, and imaginative games. Giorgia usually relies on family and familiar routines to plan activities, but expressed that safe, low-risk, and engaging suggestions would be very appealing. She had never used personalized activity apps but stated that an intuitive tool saving time could be very valuable. She values recommendations from other parents, and practical, short tips are preferred. Even during the interview it became clear that coordinating three children with different interests creates constant interruptions. This too highlights the importance of quick as well as easily digestible content.

Olena, 50, a single working mother from Gera (Germany), works full-time as a teacher and has a 10-year-old son who is very active, curious, and easily distracted. He enjoys playing videogames a lot, also drawing and playing football. But Olena finds it challenging to motivate him to learn outside school. She typically searches for new activities or learning ideas online, mainly via Google or advertisements, and values simplicity, affordability, and a clear interface. She plans her child's schedule using Google Calendar, searching for new activities seasonally. She prefers platforms offering age-specific content about education, health, and local events, with filters by age and location. She finds personalized apps useful when they provide practical solutions. Olena often exchanges resources with other parents and ensures digital safety through cautious browsing. She would use a platform like Grow & Care weekly or even daily if it addressed common parenting challenges, especially during weekends, evenings, and holidays.

Giuliana, 41, a single mother from Ferrara (Italy), works full-time and has an 8-year-old daughter who is creative, hyperactive, and easily distracted. Being an only child, she often feels lonely and spends excessive time on the tablet or watching television. Giuliana struggles to find activities that both engage her daughter and provide social interaction. She expressed interest in personalized suggestions that are educative, creative, and also offer opportunities for her daughter to meet peers. The interview showed that time constraints significantly limit her ability to plan activities, emphasizing the need for Grow&Care to provide options that are not only easy and fast to find, but also ready-to-use.

Oksana, 48, a stay-at-home mother with shared custody of three children (6, 10 and 12) from Russia, living in Germany. Her children are very different, two are hyperactive, the middle one is very calm. They often spend time on their phones, playing games with robots or legos. Oksana finds new activities online or recalls games from her own childhood, valuing safety and educational value. Her days are structured around school, appointments, and family time, planned with a calendar, alarms and notes. She seeks practical guidance on children's adaptation, socialization, and school readiness, relying mostly on her own experience and occasional pediatric advice. Her children use ANTON (german learning app for children) and other learning apps under parental control. Keeping them engaged in learning outside school is her main challenge. She would use Grow & Care regularly if it provided useful, engaging content for daily planning.

Additionally, two informal interviews were conducted with children aged 4 and 8 to learn about their preferences and behaviors during play and learning activities. The children liked colorful visuals, simple interactions, and games that offered immediate feedback or creative freedom. They showed enthusiasm for story-based and exploratory content but quickly lost interest when activities became repetitive or too challenging. This highlights the need for an intuitive, visually engaging, and varied "Child Mode" experience that keeps the children's attention while also supporting learning through play.

Across all interviews, a few common themes emerged. Parents value centralized, trustworthy activity suggestions and personalized recommendations filtered by age, location, or activity type that save time and reduce cognitive load. Safety, practicality, and ease of use are crucial, and digital tools should accommodate both planning and supervised child engagement. Opportunities for social interaction and guidance to manage screen time were particularly important for single or only-child households. Many parents draw on their own experience or tips from other parents, highlighting the value of a supportive community. Overall, these observations indicate that Grow&Care could significantly improve parental efficiency while supporting children's development and engagement.